

元智大學
YUAN ZE UNIVERSITY
管理學院財務金融暨會計碩士班
財務金融碩士學程

碩士論文計畫書
MASTER'S THESIS PROPOSAL

隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

研究生 / Student: 王新福 Christopher Ongko
指導教授 / Advisor: 孔德蓉 De-Rong Kong

中華民國 115 年 9 月
September 2026

Abstract

Digital economy is measured through several records (platform transaction value, platform revenue, national e-commerce statistics and payment data) that are often used interchangeably, although each captures a different part of the same activity. This study asks how much economic activity remains invisible when digital platforms are measured through their reported revenue. I define the invisible wedge as transaction value minus platform-recognized revenue and follow it across Indonesian platforms from 2020 to 2025. In 2023, Tokopedia, Shopee and Grab processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue. The relationship was unstable: among platforms reporting both figures, revenue grew faster than transaction value in six of eight year-to-year comparisons and moved in the opposite direction in two. Using its own definitions, BPS-Statistics Indonesia reports a 2023 marketplace figure below Tokopedia's and Shopee's transaction value alone and places 98.5 percent of 2023–2024 e-commerce growth outside its marketplace category; payment measures grew up to eleven times as fast as e-commerce value. Eight platforms outside Indonesia show the same distinction with smaller divergence. Conclusions about scale and growth therefore depend on the record used.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). That figure is one record of the activity: the gross value of the transactions involved. The platforms that carry those transactions report a different number, their own revenue; national statistical agencies report a third, the online sales of surveyed businesses; and payment systems report a fourth, the value of funds moved. These records are designed to measure different things, yet they are often used interchangeably. Substituting one for another changes the measured scale, growth and even the direction of digital activity.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as its own revenue: the commissions and fees it earns for arranging the transactions. Depending on the business model, the remaining 90 can include merchant receipts, driver payouts, inventory costs, taxes and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge. The term refers to what a revenue-only measure excludes, not to unrecorded activity. Revenue records what the platform earned, while transaction value records what it carried. The arithmetic is familiar; the measurement problem is that the relationship does not remain fixed.

Indonesia, the largest digital market in Southeast Asia, shows how far apart these records are. The study covers four marketplaces—Tokopedia, Blibli, Bukalapak and Shopee—plus Grab, whose Indonesian business is mainly delivery and other on-demand services. Tokopedia, Blibli and Bukalapak report both measures themselves. Grab's transaction value uses Indonesian revenue and its company-wide ratio; Shopee combines Momentum Works' Indonesian estimate with Sea's company-wide monetization rate. In 2023, Tokopedia, Shopee and Grab together processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion. BPS-Statistics Indonesia, the national statistical agency, placed the marketplace component of e-commerce at about US\$13.2 billion and all e-commerce at US\$72.3 billion. Only platform transaction value and revenue enter W; BPS and payment data are independent comparison records. Their definitions and scopes differ, so they are not yet directly comparable. Each answers a different question; quoting one as the size of the digital economy means choosing a particular record.

The records also disagree about growth. Between 2022 and 2023, Tokopedia's revenue rose 53.2 percent while the transaction value it processed fell 8.9 percent: read through its accounts the

platform was expanding quickly, read through its transactions it was shrinking. The reversal is not isolated. Among the three platforms that report both figures themselves, revenue grew faster than transaction value in six of eight year-to-year comparisons between 2020 and 2025, and the two moved in opposite directions twice. Nationally, BPS reports e-commerce value up 17.1 percent between 2023 and 2024, with 98.5 percent of the increase outside BPS's marketplace category, while economy-wide payment measures from Bank Indonesia, the central bank, grew between 1.8 and 11 times as fast. The wedge establishes scale; the central test is whether revenue remains a stable guide to the commerce underneath it.

A statement about how large Indonesia's digital economy is, or how fast it grew, is therefore a statement about a record. An investor using platform revenue and an analyst using marketplace figures reach different conclusions about the same years because the records measure different objects. The error lies in treating either one as the measure of the digital economy.

1.1 Research question and objectives

One question drives the proposal:

How much economic activity remains invisible when digital platforms are measured through their reported revenue?

Indonesia provides the main empirical setting. Three objectives follow. First, I measure the invisible wedge across the platform histories available from 2020 to 2025. Second, I reconcile selected large movements with disclosed changes in monetization, customer incentives, revenue recognition and reporting scope. Third, I compare the platform evidence with national e-commerce and payment records to determine whether they give the same account of economic scale and growth.

A separate comparison of eight listed platform businesses outside Indonesia, including eBay, Etsy, Shopify and Mercado Libre, tests whether the same separation between transaction value and revenue appears under other business models.

1.2 Contributions

First, the study turns the invisible wedge from a one-year estimate into a measure followed over time, and shows that revenue is not a stable guide to the commerce behind it: among the issuer-reported platforms, revenue outgrew transaction value in six of eight comparisons and moved in the opposite direction twice.

Second, I reconcile a major movement in the wedge using accounting components the platform discloses. Of the increase in Tokopedia's net revenue between 2022 and 2023, 60.6 percent reconciles to lower customer incentives and 39.4 percent to higher gross revenue, in a year when its transaction value fell.

Third, I extend the comparison beyond company accounts and show that national statistics and payment records disagree with the platform picture, and with each other, about both the level and the growth of Indonesian digital commerce.

Fourth, the study separates the figures issuers report from those that must be constructed, and shows that the longitudinal result rests on the reported figures alone.

1.3 Positioning the contribution

The contribution is a finance measurement problem. Platform reports reveal both the commerce a platform carries and the revenue it recognizes, allowing the consequences of substituting one measure for the other to be observed directly. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. This paper applies those ideas to platform economies and shows that the record chosen to describe digital activity changes the economic conclusion.

2. Literature Review and Research Gap

Digital platforms sit between economic activity and the records used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

Revenue-recognition rules affect how much of this difference appears in the accounts. Under International Financial Reporting Standard 15 (IFRS 15), whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. The literature therefore explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, the quantities are published by different producers. Industry reports size the regional digital economy by gross merchandise value (Google, Temasek and Bain 2024), national statistics measure the online sales of surveyed businesses (BPS-Statistics Indonesia 2025), and listed platforms report the revenue they recognize under IFRS 15 (International Accounting Standards Board 2014). A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. This thesis therefore compares the records rather than assuming that one can substitute for another.

2.3 Digital Records and Third-Party Information

Transactions that do not enter a platform's revenue can still leave detailed records within the platform's system. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records, yet the transactions they make on a platform are recorded by it. This differs from shadow-economy settings, where reliable formal records may be sparse (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Platform-mediated activity can also create traces outside conventional records (Berg et al. 2020; Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

Third-party information changes compliance and enforcement (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules (OECD 2020) and the European Union's DAC7 regime (European Union 2021) apply this principle to platforms, as does Indonesia's Minister of Finance Regulation No. 37 of 2025 (PMK 37/2025) through seller identification, transaction-linked reporting and marketplace withholding (Ministry of Finance of the Republic of Indonesia 2025). The relevance here is informational: what a measure or rule can show depends on the records it uses.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; official measurement frameworks explain why these quantities should not be treated as equivalent; and research on digital records explains why some platform-mediated activity outside revenue can remain observable in platform records.

What remains less clear is how large these differences become in practice, whether they persist over time, and whether choosing one record instead of another changes the economic conclusion. This thesis addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment records, and using external platform evidence to test whether the same pattern appears beyond Indonesia.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several records, but those records do not measure the same economic object. The framework begins with the difference between the value a platform processes and the revenue it recognizes, then asks whether that relationship remains stable over time and whether other records produce the same account of digital activity.

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. Figures that issuers report and figures that must be constructed are kept apart, as are national statistics and payment records (Section 4).

3.1 Measurement Framework

Let V_{it} denote transaction value processed by platform i in period t , and R_{it} the revenue recognized over the same matched period and business scope. The absolute invisible wedge is:

$$W_{it} = V_{it} - R_{it}$$

W is transaction value outside the platform's recognized revenue, not an omission from its accounts: both inputs can appear in the same company report. It is a transaction-value measure rather than an income or value-added measure; Section 7 states the corresponding inference boundaries. Because platforms differ in size, I also express the wedge relative to revenue. The Ecosystem Ratio is:

$$E_{it} = (V_{it} - R_{it}) / R_{it}$$

An E of 9 means that nine units of transaction value lie outside the revenue line for every unit recognized as platform revenue. E contains the same information as the platform's monetization rate, $t = R/V$, because $E = 1/t - 1$. For example, a platform that recognizes 2.5 percent of transaction value as revenue has an E of 39. The residual form matches the research question, which concerns commerce outside recognized revenue.

The Ecosystem Ratio is constructed for this study as a descriptive transformation of the monetization rate. Its interpretation rests on the platform-economics result that booked revenue need not track the transaction value a multisided platform coordinates (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016), together with the accounting result that the share entering revenue depends on how revenue is recognized, including whether the platform acts as principal or agent (International Accounting Standards Board 2014; De Franco, Kothari and Verdi 2011).

A single ratio does not show whether the relationship holds through time, so I define annual growth divergence, where $g(X)$ is the percentage growth of X from one year to the next, as:

$$D_{it} = g(R_{it}) - g(V_{it})$$

A positive D means revenue grew faster than transaction value, so the platform recognized a larger share of transaction value as revenue and E fell; a negative D means the reverse. If revenue were a stable proxy for transaction value, D would remain close to zero. W gives the wedge in currency units, E expresses it relative to revenue, and D carries the central test: whether the relationship between revenue and transaction value remains stable through time.

3.2 Hypotheses

The four hypotheses test the research question one record at a time. H1 concerns transaction value and revenue within platforms, H2 and H3 the national e-commerce evidence, and H4 the comparison across records. None is interpreted as a causal claim. Table 1 pairs each hypothesis with the evidence that tests it.

H1. Within the same platform, transaction value and platform revenue do not grow in proportion.

If they did, D would remain close to zero and E would be constant. The test is whether D remains close to zero; changes in its sign identify especially strong reversals but are not required for H1.

H2. National e-commerce growth comes mainly from more participating businesses rather than from higher sales per business.

Growth driven by new sellers can take place in channels that platform accounts do not cover. The split is arithmetic and does not assign new businesses to particular channels.

H3. Marketplace participation is associated with stronger financial recordkeeping.

Financial statements are one conventional record a business keeps of its own activity. H3 tests whether the prevalence of that record differs between marketplace and non-marketplace sellers; it does not treat the association as causal.

H4. Conclusions about the growth of Indonesian digital commerce change with the record used to measure it.

H4 is the central claim of the thesis. It fails if revenue grows in step with transaction value within platforms and the marketplace channel grows in step with all e-commerce. Level differences between platform and BPS figures, and payment growth, are reported as context because their definitions and coverage differ.

Table 1. Hypotheses and principal empirical tests.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Growth divergence D and the Ecosystem Ratio for the platforms that report both figures
H2 - Sources of e-commerce growth	BPS national e-commerce value split into business count and sales per business
H3 - Marketplace participation and recordkeeping	BPS published comparison of marketplace and non-marketplace sellers
H4 - Measurement choice	Growth of revenue against transaction value within platforms, and of the marketplace channel against all e-commerce in BPS

The comparison of platform businesses outside Indonesia corroborates H1 rather than forming a fifth hypothesis.

4. Data and Methodology

The study uses several records because no single one captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison. The records answer different questions and are not combined into one dataset.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity. An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified; measures referring to different periods are excluded, and major changes in accounting definitions or business structure are recorded.

The main sample consists of the platforms that report transaction value and revenue themselves: Tokopedia, Blibli and Bukalapak. Grab and Shopee, whose Indonesian figures must be constructed, support the 2023 scale comparison. Lazada and TikTok Shop, both large in Indonesia, are outside the sample because no matched Indonesian revenue series has been located for either. The platforms are disclosing cases rather than a representative sample of Indonesian platforms.

Table 2. Indonesian platform evidence.

Type of evidence	Platforms	Years and observations	Role
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	2022–2023 (2)	Main sample: cleanest scope, shortest window
Issuer-reported, broader scope	Blibli third-party business; Bukalapak Group	Blibli 2021–2025; Bukalapak 2020–2023 (9)	Main sample: most of the time series
Requires construction	Grab; Shopee Indonesia	Grab 2021–2023; Shopee 2022–2024 (6)	Supporting: 2023 scale comparison

Table 2 gives the main sample 11 platform-years and 8 year-to-year comparisons; adding Grab and Shopee gives 17 and 12. Each series covers only years for which a matched pair exists. Tokopedia's 2021 pair covers mismatched periods, and after its combination with TikTok Shop under PT Tokopedia on 31 January 2024, the listed parent reports a contractual fee rather than platform revenue. Blibli's third-party segment includes online travel; its 2020 prospectus pair is used only as a sensitivity. Bukalapak reports at Group level, including some overseas activity; its 2024 pair is excluded because it covers nine months of transactions against twelve of revenue. Grab's transaction measure changes basis after 2023.

4.2 Comparing Platform Activity Through Time

Changes through time are examined mainly by comparing each platform with itself. This keeps the company, business activity, and reporting convention as similar as the disclosures allow. When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit; these reconciliations are arithmetic rather than causal.

Companies use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). Transaction value, V, is the umbrella concept in this proposal and GTV is the label used for it, while each issuer's own label and definition are retained and checked rather than assumed to be identical.¹

Growth rates are computed in each issuer's reporting currency, so exchange-rate movements do not enter D. US dollar values are used only for the 2023 cross-section, where rupiah figures are converted at the World Bank's 2023 official exchange rate of 15,236.88 rupiah per dollar (World Bank n.d.).

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information. Table 3 sets out, for each platform, what is disclosed directly, what has to be constructed, and the limitation that follows.

Table 3. Construction of the principal platform measures.

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Indonesian ratio equals the Group ratio
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Indonesian ratio set by the Group rate
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesian goods commerce

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The Group monetization rate is platform revenue divided by gross transaction value.

Sea Limited does not report Shopee transaction value separately for Indonesia. Momentum Works supplies the Indonesian estimate; its annual Southeast Asian e-commerce report is widely cited for country-level platform estimates and is used because no issuer discloses the figure. The value is labeled as a third-party estimate wherever it appears, and Indonesian revenue is estimated using Sea's Group service monetization rate:

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate}$$

Construction fixes both Indonesian ratios to company-wide relationships: Grab's E equals its Group E, while Shopee's is the inverse of Sea's monetization rate less one. The series contribute Indonesian levels but no independent evidence on the ratio or its movement, so they remain outside the main sample.

¹ Issuer labels differ. GoTo reports GTV and Sea Limited reports GMV for Shopee; for these the label differs but the measure is the same. Blibli and Bukalapak report TPV for the segments used here, and Grab reports GMV for deliveries and mobility and TPV for financial services. Where a source reports TPV the measure rests on a different basis, so it is retained under its own label rather than relabeled as GTV.

4.4 Evidence Beyond Platform Accounts

Table 4. Records used and their role.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Measure the wedge and its movement
BPS national e-commerce statistics	Online sales of surveyed businesses, by channel, and the number of businesses	Examine the level and sources of growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Economy-wide digital payment measures	Context on payment growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

Table 4 gives each record a single role. The records answer different questions, so they are reported separately rather than pooled into one estimate.

For H2, national e-commerce value S is written as:

$$S = N \times A$$

where N is the estimated number of e-commerce businesses and A is implied nominal sales per business, and the change in S is divided arithmetically between the two. BPS publishes two counts of 2023 businesses; the count consistent with its stated growth figure is used, and the other is retained as a sensitivity check.

For H3, BPS's published business-level comparison of marketplace and non-marketplace sellers is used. For H4, BPS's split of e-commerce value by sales channel is set against the platform evidence. Bank Indonesia's economy-wide payment measures provide an independent view of digitalization. Because they cover more than e-commerce and also reflect adoption and substitution between payment methods, they are compared by growth rate and are not treated as sales estimates.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual observations for eight listed platform businesses from 2017 to 2025: eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify and Zalando. It tests whether the same separation between transaction value and revenue appears under other business models and is reported separately from the Indonesian evidence.

Robustness checks vary the sample-inclusion rules, revenue definitions, publication versions, and the assumptions in the constructed country measures. The resulting ranges are reported as sensitivity ranges rather than statistical confidence intervals.

5. Preliminary Evidence and Feasibility

The evidence assembled so far establishes that the proposed comparisons can be carried out and that the choice of record produces economically meaningful differences. The results first show the scale and movement of the wedge, then compare the platform picture with national and payment records.

5.1 Platform Scale and Movement

The 2023 comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, and Grab, a major delivery intermediary.

Table 5. Indonesia-focused 2023 platform comparison, US\$ billions.

Case	Transaction value V	Revenue R	Wedge W	Ecosystem Ratio E	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value constructed
Tokopedia	16.331	0.405	15.926	39.296x	Issuer-reported pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Three platforms	43.233	3.162	40.070	12.671x	Sum of three cases, not national total

Table 5 reports that cross-section. Across the three platforms, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized revenue, a wedge of US\$40.07 billion and a ratio of 13.7 to 1. The wedge is equivalent in magnitude to about 2.9 percent of Indonesia's 2023 nominal GDP of US\$1.371 trillion (World Bank n.d.); the comparison is of magnitude only, since the wedge is transaction value rather than value added.

Two ratios rest on construction. Shopee's 9.0 follows from an assumed company-wide monetization rate of 10 percent; Grab's 7.9 is its company-wide ratio. Neither provides independent Indonesian ratio evidence. Varying Shopee's rate from 9 to 11 percent puts the combined wedge between US\$39.86 billion and US\$40.29 billion; excluding any platform leaves at least US\$20.70 billion.

At the 2023 exchange rate, Blibli's and Bukalapak's reported transaction and revenue figures yield wedges of US\$3.20 billion and US\$10.50 billion. They are excluded from the total because their measures include online travel and Group-level activity. Their ratios, 43.4 and 36.0, sit close to Tokopedia's 39.3; the constructed ratios instead reflect company-wide monetization.

The relationship is not stable. Revenue grew faster in six of the eight main-sample comparisons, and the median absolute gap between the two growth rates was 38.92 percentage points. In two comparisons, revenue and transaction value moved in opposite directions. Tokopedia's 53.2 percent revenue growth against an 8.9 percent transaction-value decline gives a 62.1-point divergence. After dropping the largest comparison, revenue still grew faster in five of seven, with a median gap of 15.74 points. Every leave-one-out sample retains both a majority and at least one reversal. Adding Grab and Shopee gives ten of twelve, but those four comparisons track company-wide monetization. Figure 1 traces E for each platform; every series ends below its starting point, although the decline is not monotonic in every case.

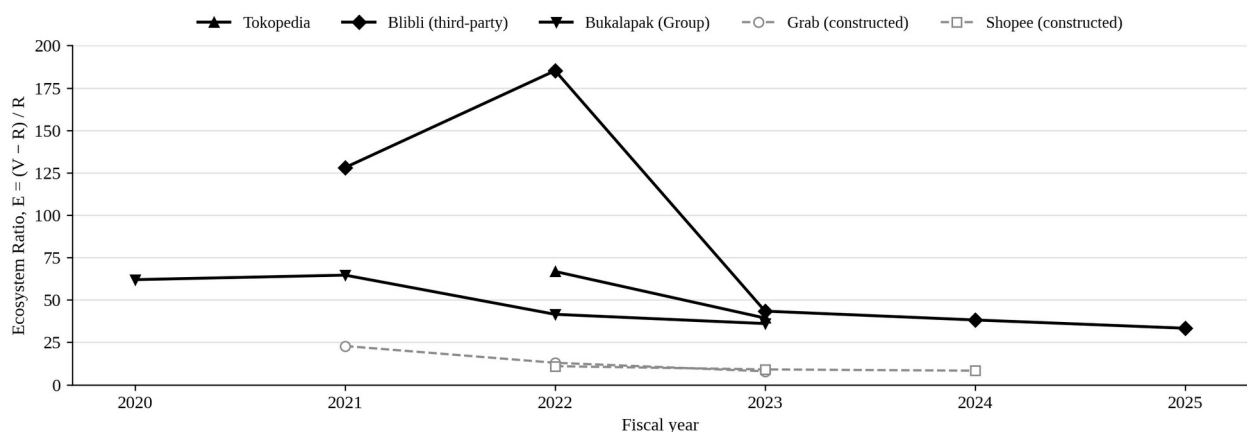


Figure 1. Ecosystem Ratio by platform, 2020–2025. Solid lines are issuer-reported pairs; dashed lines are constructed Indonesian series whose ratios equal Group ratios by construction.

Tokopedia illustrates the separation. In 2022–2023, transaction value fell 8.9 percent while third-party net revenue rose 53.2 percent; 60.6 percent of the revenue increase reconciles to lower customer incentives and 39.4 percent to higher gross revenue. Its accounts show growth while its

transactions show contraction. Blibli similarly reversed in 2024–2025: transaction value fell 1.89 percent and net revenue rose 12.07 percent. The pattern therefore appears in more than one platform and period.

5.2 Evidence Beyond Platform Revenue

National statistics give a different account of the same commerce, beginning with its level. BPS measures online sales reported by surveyed e-commerce businesses; Tokopedia reports GTV, while the external Shopee figure is a GMV estimate. For 2023, BPS places the marketplace component at Rp200.68 trillion, about US\$13.2 billion, in its exclusive channel split, where marketplaces account for 18.2 percent of e-commerce value. Tokopedia's reported GTV and the external Shopee estimate total US\$37.9 billion. Because channel attribution and reporting definitions differ, the records are not yet like-for-like. The thesis will reconcile those definitions before attributing any part of the difference to coverage.

Growth can be compared more directly. Table 6 summarizes the evidence for H2–H4 and the payment context.

Table 6. Preliminary evidence beyond platform accounts.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; sales per business +1.54%. The business count accounts for 90.29% of the increase (70.95% under BPS's other count)	Growth comes mainly from more businesses selling online (H2)
Where does that growth occur?	Marketplace channel +1.45%; other channels +20.57%; 98.46% of the increase is outside the marketplace channel	Most measured growth occurs outside BPS's marketplace category (H4)
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger recordkeeping (H3)
Do payment records show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; Quick Response Code Indonesian Standard (QRIS) +186.98%	Context only: payment measures grow 1.8 to 11 times as fast, and also reflect adoption and substitution

BPS also reports widespread use of non-marketplace channels. In 2023, 95.33 percent of e-commerce businesses used instant messaging. In its multiple-response measure, where a business can report several channels, social media accounted for 44.04 percent of transaction value and marketplaces for 32.74 percent; that measure is not comparable with the exclusive 18.2 percent split above. These figures describe the 2023 channel structure rather than the allocation of 2023–2024 growth. The business-count split is arithmetic and the recordkeeping result is an association.

5.3 Feasibility and Scope of the Preliminary Findings

The preliminary evidence supports each hypothesis: H1 through the six of eight comparisons in which revenue outgrew transaction value, H2 through the business-count share of growth, H3 through BPS's business-level comparison, and H4 through the different growth recorded within platforms and within the national channel split.

Outside Indonesia, transaction value exceeds revenue in all eight businesses. In the 29 comparisons not affected by acquisitions or changes in reporting scope, revenue grew faster than transaction value in 22 and the measures moved in opposite directions in 4. The median gap is 7.1 percentage points, compared with 38.92 points in the Indonesian main sample. Among the marketplaces, monetization rates range from about 2.4 percent at Shopify to about 25 percent at Mercado Libre.

PMK 37/2025 provides an administrative example. From 1 November 2026, the Directorate General of Taxes (DJP) requires designated marketplace operators to report seller-linked transactions and withhold tax. Those operators cannot yet be mapped directly to BPS's marketplace category; compliance and revenue effects remain outside this study.

6. Why the Problem Matters

Indonesia's digital economy is observed through platform accounts, transaction measures, national statistics and payment records, which disagree about scale and growth. Tokopedia's transaction value fell 8.9 percent while its revenue rose 53.2 percent, and BPS's marketplace category grew 1.45 percent while all e-commerce grew 17.08 percent. The contribution is to identify which conclusions depend on the record selected and why no fixed conversion from platform revenue can recover the commerce underneath it.

7. Limitations and Robustness

The evidence is descriptive. The main sample has three platforms, 11 platform-years and 8 comparisons, supporting description rather than statistical inference. Scope differs: Tokopedia is observed for two years, Blibli includes online travel, and Bukalapak reports at Group level. Grab and Shopee are constructed, their ratios contain no independent Indonesian information, and Shopee's revenue uses an assumed monetization rate. Section 5.2 also compares differently defined records whose disagreement is measured but not yet reconciled.

Table 7 pairs each supported conclusion with what it does not establish.

Table 7. Main inference boundaries.

Evidence supports	Does not by itself establish
W, E, and D describe the size and movement of the wedge.	Participant profit, taxable income, unpaid tax, tax evasion, or missing GDP.
Issuer-reported pairs show revenue and transaction value moving apart within platforms.	Independent Indonesian ratio evidence from the constructed Grab and Shopee series.
Disclosed revenue components reconcile selected movements in the wedge.	A causal effect of incentives, monetization, or accounting choices.
BPS evidence describes the sources and channels of e-commerce growth and recordkeeping associations.	Causal business entry, productivity effects, or marketplace-induced formalization.
Platform, BPS and payment records disagree in level and growth.	Which record, if any, contains measurement error, or that payment values measure sales.
PMK 37/2025 draws on designated operators' transaction records.	Its coverage of the digital economy, or a compliance or tax-revenue effect.

Robustness is reported as sensitivity ranges: Shopee's assumed rate is varied, each platform is excluded in turn, the largest comparison is dropped, and the longitudinal result is restricted to issuer-reported pairs.

8. Next Steps

The data and empirical framework are assembled. The remaining work will reconcile the definitions behind the BPS and platform levels, divide movements in the wedge into transaction-volume and monetization effects, test alternative sample-inclusion rules, and complete the robustness checks. Table 8 sets out that schedule.

Table 8. Planned work toward the final thesis.

Period	Planned work
Proposal stage	Incorporate examination feedback; finalize the treatment of broader-scope and constructed series.
October–November 2026	Reconcile BPS and platform definitions; divide wedge movements into volume and monetization effects; complete robustness checks; finalize tables and figures.
Before the final defense	Complete the full manuscript and the final citation and consistency review.

The thesis treats the choice of record as an empirical question. It measures how far the records disagree, tests whether the relationship between revenue and transaction value remains stable, and identifies which conclusions about the size, growth and direction of digital activity depend on the record chosen.

Appendix A. Variables and Data Sources

Table A1. Variable definitions and sources.

Variable	Definition	Source
V	Transaction value processed through the platform in the period	Issuer disclosure, or construction (Table 3)
R	Revenue the platform recognizes for the same period and business scope	Issuer disclosure, or construction (Table 3)
$t = R / V$	Monetization rate: the share of transaction value recognized as revenue	Computed
$W = V - R$	Invisible wedge: transaction value outside recognized revenue	Computed
$E = (V - R) / R = 1/t - 1$	Ecosystem Ratio: wedge per unit of revenue; constructed for this study	Computed
$D = g(R) - g(V)$	Growth divergence: revenue growth minus transaction-value growth	Computed from consecutive years
$S = N \times A$	National e-commerce value: number of businesses times sales per business	BPS-Statistics Indonesia

Table A1 defines each variable and says where it comes from. Table A2 traces each record to the filings and publications behind it.

Table A2. Source lineage by evidence type.

Evidence type	Platform / source	Transaction-value source	Revenue / comparison source
Issuer-reported, Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Issuer-reported, broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Issuer-reported, broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Requires construction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Requires construction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count, channel and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
Exchange rate and GDP	World Bank	World Development Indicators, 2023	Converts 2023 rupiah values; GDP scale comparison
External corroboration	eBay, Etsy, Jumia, Mercado Libre, Rakuten, Sea, Shopify, Zalando	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025; DJP	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Every figure used in this proposal traces to a named file and locator in the accompanying data package.

References

- Ahmad, N., & Schreyer, P. (2016). Measuring GDP in a Digitalised Economy. OECD Statistics Working Papers 2016/07.
- Armstrong, M. (2006). Competition in two-sided markets. *RAND Journal of Economics*, 37(3), 668-691.
- Bank Indonesia. (2025). Payment System Statistics (SPIP) and QRIS reports. Jakarta: Bank Indonesia.
- Barrios, J. M., Hochberg, Y. V., & Yi, H. (2022). Launching with a parachute: The gig economy and new business formation. *Journal of Financial Economics*, 144(1), 22-43.
- Berg, T., Burg, V., Gombovic, A., & Puri, M. (2020). On the rise of FinTechs: Credit scoring using digital footprints. *Review of Financial Studies*, 33(7), 2845-2897.
- BPS-Statistics Indonesia. (2025). E-Commerce Statistics 2023 and 2024. Jakarta: BPS.
- Caillaud, B., & Jullien, B. (2003). Chicken and egg: Competition among intermediation service providers. *RAND Journal of Economics*, 34(2), 309-328.
- De Franco, G., Kothari, S. P., & Verdi, R. S. (2011). The benefits of financial statement comparability. *Journal of Accounting Research*, 49(4), 895-931.
- Denes, M. R., Lagaras, S., & Tsoutsoura, M. (2025). Entrepreneurship and the gig economy: Evidence from U.S. tax returns. *Journal of Financial Economics*, 173, 104156.
- European Union. (2021). Council Directive (EU) 2021/514 amending Directive 2011/16/EU (DAC7).
- Evans, D. S., & Schmalensee, R. (2016). *Matchmakers: The New Economics of Multisided Platforms*. Boston: Harvard Business Review Press.
- Google, Temasek, & Bain & Company. (2020). e-Conomy SEA 2020: At Full Velocity.
- Google, Temasek, & Bain & Company. (2024). e-Conomy SEA 2024: Profits on the Rise.
- Hagiu, A., & Wright, J. (2015). Multi-sided platforms. *International Journal of Industrial Organization*, 43, 162-174.
- International Accounting Standards Board. (2014). IFRS 15: Revenue from Contracts with Customers.
- International Monetary Fund. (2018). Measuring the Digital Economy. IMF Policy Paper.
- Kleven, H. J., Knudsen, M. B., Kreiner, C. T., Pedersen, S., & Saez, E. (2011). Unwilling or unable to cheat? Evidence from a tax audit experiment in Denmark. *Econometrica*, 79(3), 651-692.
- Kleven, H. J., Kreiner, C. T., & Saez, E. (2016). Why can modern governments tax so much? *Economica*, 83(330), 219-246.
- La Porta, R., & Shleifer, A. (2014). Informality and development. *Journal of Economic Perspectives*, 28(3), 109-126.
- Medina, L., & Schneider, F. (2019). Shedding light on the shadow economy: A global database. CESifo Working Paper 7981.
- Ministry of Finance of the Republic of Indonesia. (2025). PMK 37 Tahun 2025; Directorate General of Taxes marketplace guidance.
- Momentum Works. (2024). Ecommerce in Southeast Asia. Singapore: Momentum Works.
- Naritomi, J. (2019). Consumers as tax auditors. *American Economic Review*, 109(9), 3031-3072.
- OECD. (2020). Model Rules for Reporting by Platform Operators with respect to Sellers in the Sharing and Gig Economy.
- OECD. (2023). OECD Handbook on Compiling Digital Supply and Use Tables. Paris: OECD Publishing.
- Parker, G. G., & Van Alstyne, M. W. (2005). Two-sided network effects: A theory of information product design. *Management Science*, 51(10), 1494-1504.
- Pomeranz, D. (2015). No taxation without information: Deterrence and self-enforcement in the value added tax. *American Economic Review*, 105(8), 2539-2569.
- Rochet, J.-C., & Tirole, J. (2003). Platform competition in two-sided markets. *Journal of the European Economic Association*, 1(4), 990-1029.
- Slemrod, J. (2019). Tax compliance and enforcement. *Journal of Economic Literature*, 57(4), 904-954.
- Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, 108(8), 2015-2047.
- United Nations et al. (2025). System of National Accounts 2025.
- World Bank. (n.d.). World Development Indicators: GDP (current US\$) and official exchange rate (LCU per US\$, period average), Indonesia. Washington, DC: World Bank.